

Enhancing microstructural integrity and mechanical strength of mortar containing incinerated ash using carbon nanotube, graphene nanoplatelet and nano silica reinforcements

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ABSTRACT

Incineration ash contains metallic aluminium that can react with cement alkalis, producing hydrogen gas, which can affect the strength of cementitious materials. In this study, we aimed to improve the technical properties of incinerated ash-based mortar using three different nanomaterials: carbon nanotubes, graphene nanoplatelets, and nano-silica. These nanomaterials were incorporated at concentrations of 0.025 wt%, 0.05 wt%, 0.1 wt%, 0.25 wt%, 0.5 wt%, and 1 wt% of the cement mass, respectively. Comprehensive analysis indicates that the type of nanomaterials and their doses play a significant role in early age hydration, shortening the induction period and promoting the formation of C-S-H. Study results show evidence up to a 45 % enhancement in compressive strength. Sustainability assessment reveals that a dose of 0.025 % graphene nanoplatelets is the most sustainable from an environmental perspective, with approximately a 25 % improvement in compressive strength of incinerated ash mortar. Findings of the study will contribute to the understanding of nanomaterial-reinforced cementitious systems incorporating incinerated ash and offer valuable guidance for improving the performance of incinerated ash-based cementitious materials in construction applications.

1. Introduction

Concrete, the most widely used composite material in construction, is highly valued for its exceptional compressive strength, fire resistance, and durability. However, as the second most consumed material after water, concrete contributes significantly to the depletion of natural rocks and sand. Consequently, exploring the potential waste materials to replace natural aggregates or binder is crucial [1]. Hence, the use of incinerated ash can solve dual problems: providing a source of aggregate and addressing waste management issues. However, due to the mineralogical composition, cementitious materials incorporating incinerated ash might achieve weaker mechanical performance compared to natural sand. Hence, improving the performance of cementitious systems with incinerated ash is crucial for achieving higher mechanical strength and expanding its application.

In recent years, researchers have used different types of fibres and nanomaterials in cementitious systems to improve mechanical, microstructural, and durability performances [2]. Nanomaterials are not only

used for performance improvement of cementitious materials, but they are also used in alkali-activated materials for performance improvement [3–6]. These materials include carbon nanotubes [5], graphene nanoplatelets [7], nano silica [8], nano-alumina [6], carbon dots [9] and others. Nano silica is widely used in cementitious systems, while new-generation carbon-based nanomaterials like carbon nanotube (CNT) and graphene nanoplatelet (GNP) are also gaining popularity due to their effectiveness in improving the technical properties of concrete with just a small dose of nanoparticles. CNTs are nano-scale carbon-based materials characterized by tubular structures formed from graphene sheets rolled up in a sp^2 configuration [10]. GNPs are flat sheets of graphene arranged in a platelet shape and stacked atop each other [11]. While nano silica (NS) known as quartz dust or silica dust, shares similarities with silica fume in its composition, with a SiO_2 content exceeding 99 % [12]. The effectiveness of nanomaterial reinforcement relies on various parameters, including the aspect ratio and surface area of nanomaterials, dispersion of nanomaterials, and their interaction with cementitious materials [13].

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Incorporation of nanomaterials in cementitious systems plays crucial roles in enhancing the mechanical, durability, and microstructural properties by providing filling, reinforcing, and nucleation effects [14, 15]. The nucleation effect of nanomaterials leads to an increase in the formation of calcium silicate hydrate (C-S-H) and the formation of denser CSH gel [16]. Incorporation of nanomaterials can also significantly accelerate the heat of hydration of cement and improve the interfacial transition zone between aggregate and cementitious materials, resulting in improved technical properties [17,18]. Wang et al. [19] reported around a 24 % improvement in compressive strength by adding 0.5 % graphene nanoparticles to cement composites. Ranjbar et al. observed a 1.44 times improvement in compressive and flexural strength of geopolymer concrete by adding graphene nanoparticles [20]. Rashid et al. reported a 269 % increase in flexural strength of CNTs reinforced cementitious materials. Similarly, a study reported that the use of 4 % nano-Fe₂O₃ in self-compacting concrete resulted in a 93.75 % improvement in mechanical performance [21]. Along with the enhancement in mechanical performance, researchers have also reported improvements in the durability performance of nanomodified cementitious materials. Julián et al. [22] reported a 70 % improvement in chloride migration coefficients by incorporating nano silica. Hawreen et al. [23] have reported a 15 % and 18 % reduction in shrinkage and creep of concrete with CNT. Similarly, many researchers have observed improvements in water absorption and permeable voids in cementitious systems incorporating nanomaterials [24–26]. The evident improvements in technical performance of cementitious materials have resulted in the gaining popularity of the use of nanomaterials in the construction industry.

Literature studies have suggested promising benefits of nanomaterials, and they have been used in concrete, cement paste, high-performance concrete, etc. The relative increase in microstructure, mechanical and durability performance of nano-modified cementitious materials depends upon various factors, including mixing composition, used materials, doses of nano materials, and type of nanomaterials. Therefore, it is crucial to explore the impacts of nano-modified cementitious systems incorporating various materials to understand the performance of nanomaterials in different cementitious systems. Literature studies suggest that the impact of nanomaterials on incinerated ash-based cementitious systems is still untouched. Incinerated ash contains considerable amounts of metallic aluminium that can react with alkalis present in cement, resulting in the formation of hydrogen gas [27]. The generation of hydrogen gas causes expansion and the formation of pores and cracks in the cementitious system, weakening the performance of cementitious materials. Therefore, the performance efficiency of nanomaterials on pre-filling and strength can differ in nano-modified cement mortar with incinerated ash compared to nano-modified cement mortar with natural sand or other aggregates. Furthermore, the comparative study of the effectiveness of different nanomaterials on different cementitious systems incorporating industrial side streams has not been fully explored either. In this study, we investigated the effectiveness of three different nanomaterials, including carbon nanotubes, GNPs, and NS, on the mechanical and microstructural properties of incinerated ash mortars at concentrations of 0.025 wt%, 0.05 wt%, 0.1 wt%, 0.25 wt%, 0.5 wt%, and 1 wt% of cement mass, respectively. In this study, isothermal calorimetry, optical microstructure analysis, scanning electron microscope imaging, water absorption testing, compressive strength testing, heavy metal leaching, thermography, FTIR analysis, and XRD analysis were conducted. The findings of the study provide an in-depth analysis of the effects of carbon and inorganic nanoparticles and their concentrations on the performance of incinerated ash-based cement mortars. This study will pave the way for the exploration of incinerated ash-based cementitious materials and provide valuable references for enhancing performance and addressing issues with incinerated ash-based cementitious systems.

2. Materials and methods

2.1. Materials

Bottom ash from co-combustion of solid recovered fuel and biomass (wood and peat) were obtained from a powerplant in Oulu, Finland (Oulun Energia). CEM II 42.5 N was used as binder and 0–2 mm particle size bottom ash was used as fine aggregate. The bulk density of the bottom ash was 1408 kg/m³ with surface areas of 0.28 [m²/g]. The chemical composition of the ashes was analysed using X-ray fluorescence spectroscopy (PanAnalytical Minipal 4). Silicon (as SiO₂) content in the bottom ash and fly ash was around 64 %. The chemical composition and physical properties of bottom ash presented in Table 1. XRD of the ashes presented in the Fig. 1 indicate the presence of quartz and anorthite in bottom ash. In this study, three different type of nano materials, multiwalled CNTs (3–5 nm inner diameter, outer diameter of 8–15 nm, lengths of 3–12 µm, specific surface area > 233 m²/g), GNPs (thickness ~2 nm, specific surface area 750, purity 99.5 %), and NS (5–15 nm, 99.5 % trace metal basis) was used at a concentrations of 0.025 wt%, 0.05 wt%, 0.1 wt%, 0.25 wt%, 0.5 wt%, and 1 wt% of cement mass, respectively. Polycarboxylate ether-based superplasticizer was used for dispersion of nano particles.

Fig. 2 shows the Transmission Electron Microscopy (TEM) image of nanomaterials, suggesting the presence of nanoparticles in agglomerated form. This agglomeration of nanoparticles suggests the need for proper dispersion methods. Fig. 2b clearly indicates that the Carbon Nanotubes (CNTs) are indeed multiwalled and have a higher length compared to the GNPs and NS. The TEM image also confirms the tube-shaped carbon nanotubes and the spherical shape of the NS. However, due to serious agglomeration, the proper shape of the GNPs wasn't identified.

2.2. Sample preparations and dispersion of nano material

The mixing composition of the nano-reinforced incinerated ash mortar is presented in Table 2. The water/binder ratio was kept at 0.4 for all samples. In the first stage of sample preparation, nano materials and water mixed with superplasticizer were poured into a container and left on a shaking table for 24 hours to disperse the nano materials. During the mixing process all the dry materials, including cement and bottom ash, were mixed in a container for 2 minutes using a mechanical mixer at 300 rpm. Subsequently, the solution of dispersed nano materials was mixed with the dry mixes and further mixed for another

Table 1

Chemical composition and physical properties of bottom ash derived from co-combustion of solid recovered fuel and biomass (wood and peat).

Chemical composition	Bottom ash, %
Na ₂ O	2.88
MgO	2.02
Al ₂ O ₃	9.75
SiO ₂	64.20
P ₂ O ₅	0.68
SO ₃	0.22
Cl	0.06
K ₂ O	3.48
TiO ₂	0.67
CaO	10.01
Fe ₂ O ₃	3.65
Others	2.41
Bulk density	1408.8 kg/m ³
Free lime content	0.10 %
Loss on ignition at 950 °C	0.16 %
Reactivity to bound water	49.90 %
Specific surface area	0.28 m ² /g
Total organic carbon	< 0.5 %
Metallic Al ⁰	0.026 %

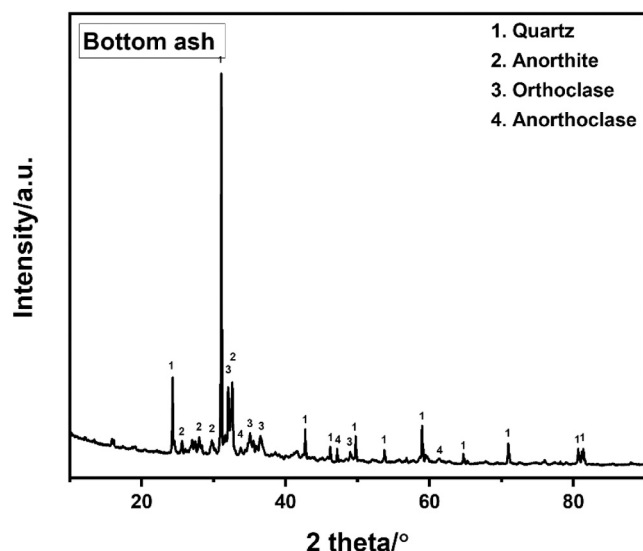


Fig. 1. XRD of bottom ash.

2 minutes. Fresh mortars were then taken out from the mixer, and the workability of the mortar was measured before proceeding with moulding the sample into 20*20*80 mm prisms. After 24 hours of the hardening process, samples were demoulded and placed in a humidity chamber for 28 days of hydration. The systematic view of nanomaterial dispersion and sample preparation is presented in Fig. 3.

2.3. Characterization techniques

The morphologies of CNTs, GNPs, and NSs are detected with an energy-filtered transmission electron microscope equipped with a scan generator (EFTEM/STEM) (Jeol JEM-2200FS). The samples are sonicated in isopropanol for 5 min and then the dispersion is dropped on a copper grid for analysis.

Isothermal analysis of samples was conducted using a TAM Air isothermal calorimeter (TA Instruments, USA) to study the heat produced and cumulative heat release during the hydration of mortars incorporating different nanomaterials. 15 g of fresh mortar samples were transferred into ampoules and placed in the calorimetry instrument. Ampoules filled with deionized water were used as references, maintaining a constant temperature of 23°C.

The compressive strength of the samples was measured after 28 days of hydration using Zwick testing equipment, with a maximum load of 50 kN and a loading force of 2.4 kN/s. The average value of three specimens from each sample was considered the result.

The surface morphology of mortar samples was analysed using a Zeiss Ultra plus instrument, equipped with a field-emission scanning electron microscope and an energy dispersive X-ray spectrometer (FE-SEM-EDS). Prior to SEM analysis, carbon coating was applied to the AAM samples.

The X-ray diffraction analysis was performed with X-ray diffractometer manufactured by Bruker with slow scanning of 0.5–2 degrees per minute.

After 28 days, samples were taken out of the water and kept at room temperature for another 14 days before undergoing thermogravimetric analysis. Simultaneous thermal analysis was performed using a thermal analyser, the STAR Systems. This instrument is utilized to study the thermal behaviours and phase transitions in materials by measuring weight changes and heat flow as a function of temperature, employing a slow scanning rate of 1–20°C/minute.

The functional groups and chemical bond in PC and nanomaterial mixed mortar specimens are investigated by using Fourier-transform infrared (FTIR) spectroscopy (Bruker Vertex 80). The spectra are

measured at room temperature with a N2 cooled MCT detector and DRIFT (diffuse reflectance infrared Fourier transform) cell accessory. The range of wavenumber is from 400 to 4400 cm^{-1} .

The leaching of heavy metals from the specimens was conducted following the EN 12457–2:2004 standard, with samples being crushed to ensure a size smaller than 4 mm. In this investigation, we deliberately crushed the samples prior to testing to study an end-of-life scenario for the developed binders. The leaching test was conducted only on the nano-modified incinerated ash mortar samples achieving optimum strength at their respective doses.

The environmental sustainability assessment is carried out considering the system boundary as shown in Fig. 3. However, it is important to mention here that the energy for the mortar preparation is the same for all dosages of nanomaterials, therefore is not considered in the assessment and only the production of the nanomaterials is considered in the system boundary as cradle to gate. Due to the lack of material datasheets from the manufacturers of CNT, GNP, and NS used in the present study, the energy and mass balance data were obtained from the literature. The calculation of inputs from nature and the technosphere are taken from [1,2] for CNT, [3,4] for GNP, and [5] for NS. The assessment is carried out using the European CML-IA baseline method [6] using the commercial SimaPro 9.0® [7] software package using the Ecoinvent 3.0 [8] inventory.

3. Results and discussion

3.1. Heat of hydration

The type of nanomaterials and their doses significantly influence the produced heat and cumulative heat of the samples. Fig. 4 shows the heat evolution and heat released by mortar samples with different nanomaterials at varying doses. It was observed that increases in nanomaterials accelerate the hydration process, shortening the induction period. This acceleration in the induction period with a higher concentration of nanomaterials was observed for all the nanomaterials. However, as Fig. 4 shows, mortar with 1 % CNT exhibits a shorter induction period compared to the other nanomaterials. Fig. 4 also evidences that the second peak of the hydration curve, called silicate, is associated with rapid C3S hydration and the formation of C-S-H. With the increase in CNT and NS doses, the hydration intensity was observed to increase, evidencing the C-S-H formation kinetics of nanomaterials. This acceleration might be attributed to the nucleation effect of nano particles [17,28,29]. In addition, it was observed that the silicate reaction was accelerated with the increase in CNT and NS doses. A similar accelerated silicate reaction was noticed for GNPs. However, a slight drop in intensity in the silicate reaction was observed when the GNP dose was above 0.025 %. This might be explained as GNP having an optimal dose of 0.025 % for accelerated silicate hydration and achieving high reaction intensity, as observed in previous study [28]. The third peak shown in Fig. 4 corresponds to the rapid release of aluminate phases from the binder. Study results indicate that the nanomaterials also impacted the hydration reaction of the aluminate phases. It was observed that mortar incorporating CNTs and NS exhibits a comparatively higher intensity of hydration reaction and accelerated hydration of aluminate phases compared to GNPs. [10,12,39]

The exothermic heat release of the mortar with CNTs and NS increases with the increase in their doses. A mortar sample with 1 % CNT gains a 39.8 % increase in heat flow and a 7.5 % rise in cumulative heat over the mortar sample with 0.025 % CNT. However, the total heat release of mortar above 0.25 % becomes almost the same. While with GNP, accelerated hydration was observed, but the produced heat slightly dropped with an increase in GNP concentrations. The total heat release of mortar samples with also increases with the increase in doses of NS; around a 38.8 % increase in heat released was observed for the sample with 1 % NS over the sample with 0.025 % NS. Comparing the cumulative heat release of different nanomaterials, NS incorporated

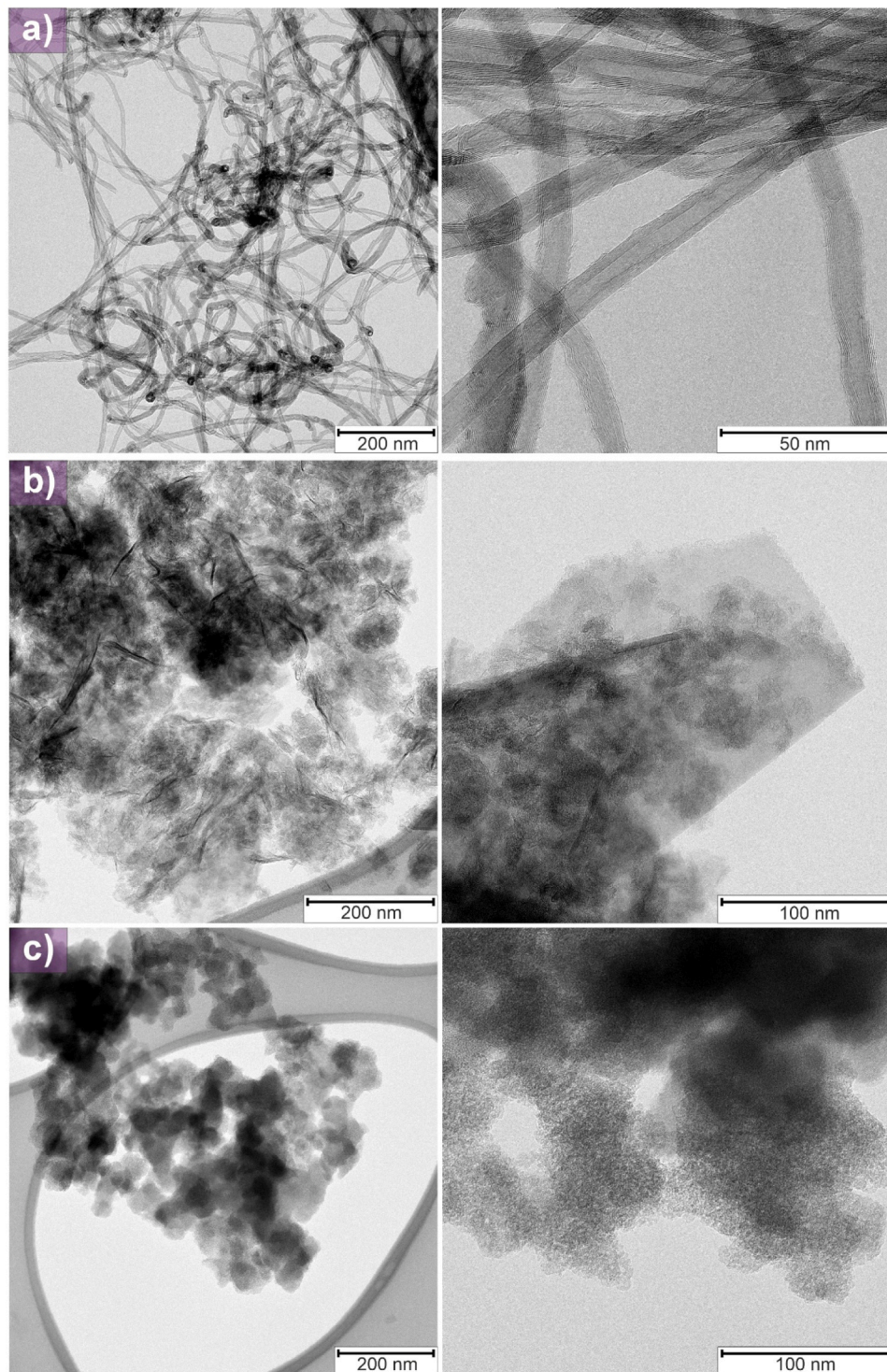


Fig. 2. TEM of carbon nanotubes, graphene nanoplatelets, and nano silica.

mortar gain maximum exothermic heat release over the samples with CNTs. The heat release of the mortar is mainly associated with the hydration of C3S [30]. The acceleration of heat release indicates the accelerated hydration of C3S and the formation of higher CSH, attributed to the nucleation effects of nano materials [31]. The heat release of mortar with NS at 0.25 % doses was observed to be reduced compared to the sample with 0.025 % NS, and certainly increased again with 1 % NS. This phenomenon suggests the agglomeration of nanoparticles and improper dispersion of nanomaterials, resulting in hindering the optimal positive effect of nanomaterials [32]. The SEM image presented in

Section 3.3 evident agglomeration of nano particles. Researchers have reported that ultrasonic dispersion and high shear mixing are efficient and commonly used dispersion methods [33]. However, the agglomeration of nanoparticles might also depend on the geometry of nanomaterials, dispersion time, and energy [34].

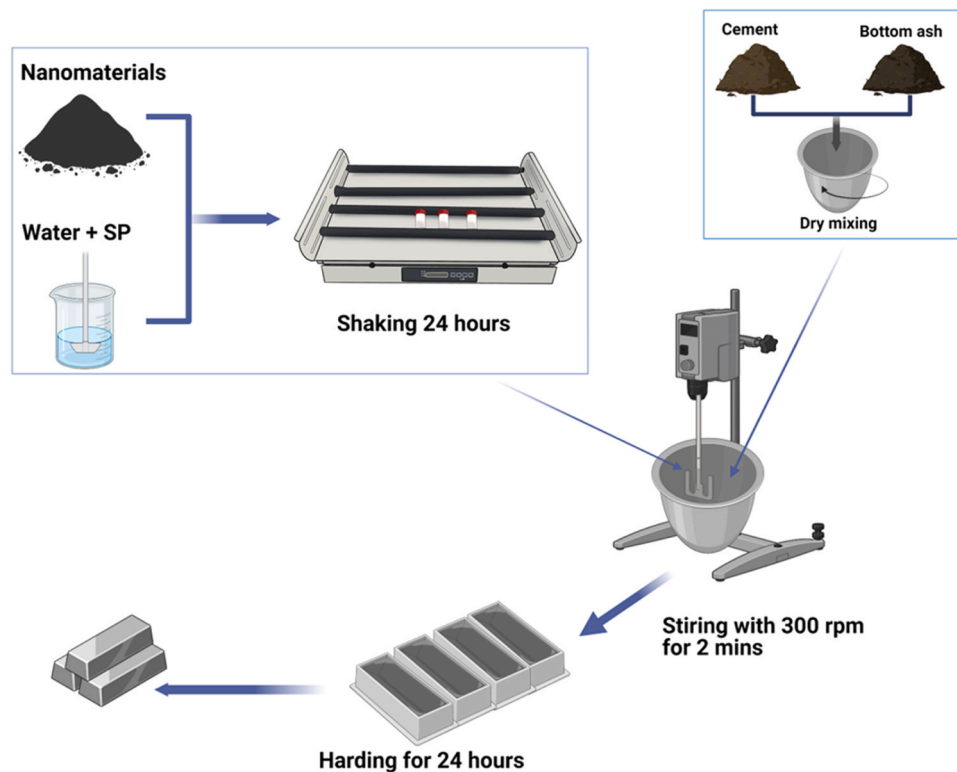
3.2. Compressive strength

Fig. 5 evident that the type of nano materials and their doses have significant impacts on the strength development of incinerated ash-

Table 2

Mix composition of nano modified incinerated ash mortar.

sample name	Cement, g	Bottom ash, g	Nanomaterials			Water		Superplasticizer	
			Type	%	g	g	W/B	g	%
Control sample	300	400	-	-	-	120	0.4	3	1 %
CNT0.025	300	400	CNT	0.025	0.075	120	0.4	3	1 %
CNT0.05	300	400	CNT	0.05	0.15	120	0.4	3	1 %
CNT0.1	300	400	CNT	0.1	0.3	120	0.4	3	1 %
CNT0.25	300	400	CNT	0.25	0.75	120	0.4	3	1 %
CN0.5	300	400	CNT	0.5	1.5	120	0.4	3	1 %
CNT1	300	400	CNT	1	3	120	0.4	3	1 %
GNP0.025	300	400	GNP	0.025	0.075	120	0.4	3	1 %
GNP0.05	300	400	GNP	0.05	0.15	120	0.4	3	1 %
GNP0.1	300	400	GNP	0.1	0.3	120	0.4	3	1 %
GNP0.25	300	400	GNP	0.25	0.75	120	0.4	3	1 %
GNP0.5	300	400	GNP	0.5	1.5	120	0.4	3	1 %
GNP1	300	400	GNP	1	3	120	0.4	3	1 %
NS0.025	300	400	NS	0.025	0.075	120	0.4	3	1 %
NS0.05	300	400	NS	0.05	0.15	120	0.4	3	1 %
NS0.1	300	400	NS	0.1	0.3	120	0.4	3	1 %
NS0.25	300	400	NS	0.25	0.75	120	0.4	3	1 %
NS0.5	300	400	NS	0.5	1.5	120	0.4	3	1 %
NS1	300	400	NS	1	3	120	0.4	3	1 %

**Fig. 3.** systematic view of nano material dispersion and sample preparation.

based mortar. Mortar samples with CNTs exhibit optimum strength at 0.25 % doses, afterward, strength starts dropping. Similarly, GNP-modified mortar samples exhibit their optimal strength at 0.025 % doses. While the compressive strength of the mortar increases with the rise in NS content, no drop in strength was measured. The relative increase in compressive strength of mortar with CNT, GNP, and NS at their optimum doses was measured at 29 %, 24 %, and 45 % respectively. Compared to the effectiveness in improving compressive strength, NS performs better than CNTs and GNPs. However, when considering the effectiveness at low doses of nanomaterials, GNPs and CNTs work better than NS. At a dose of 0.025 %, GNP improves the compressive strength by almost 25 %, while at a dose of 0.25 %, CNTs improve the

compressive strength by around 30 %, which is greater than the improvement provided by NS. The enhancement in strength might be attributed to the nucleation of nano materials, resulting formation of higher content of CSH [5]. While the drop in mechanical performance after certain doses of CNTs and graphene nano platelets might be attributed to the agglomerations of nanoparticles [5,35,36]. While the higher strength development of mortar samples with NS might be attributed to the formation of higher content of CSH, attributed to the supply of additional silica. Calorimetry results also evidence higher heat release and cumulative heat. Despite the higher cumulative heat of GNP mortar over CNT, the drop in strength above 0.025 % might be attributed to the surface geometry of GNP [28]. Konsta-Gdoutos et al. [37]

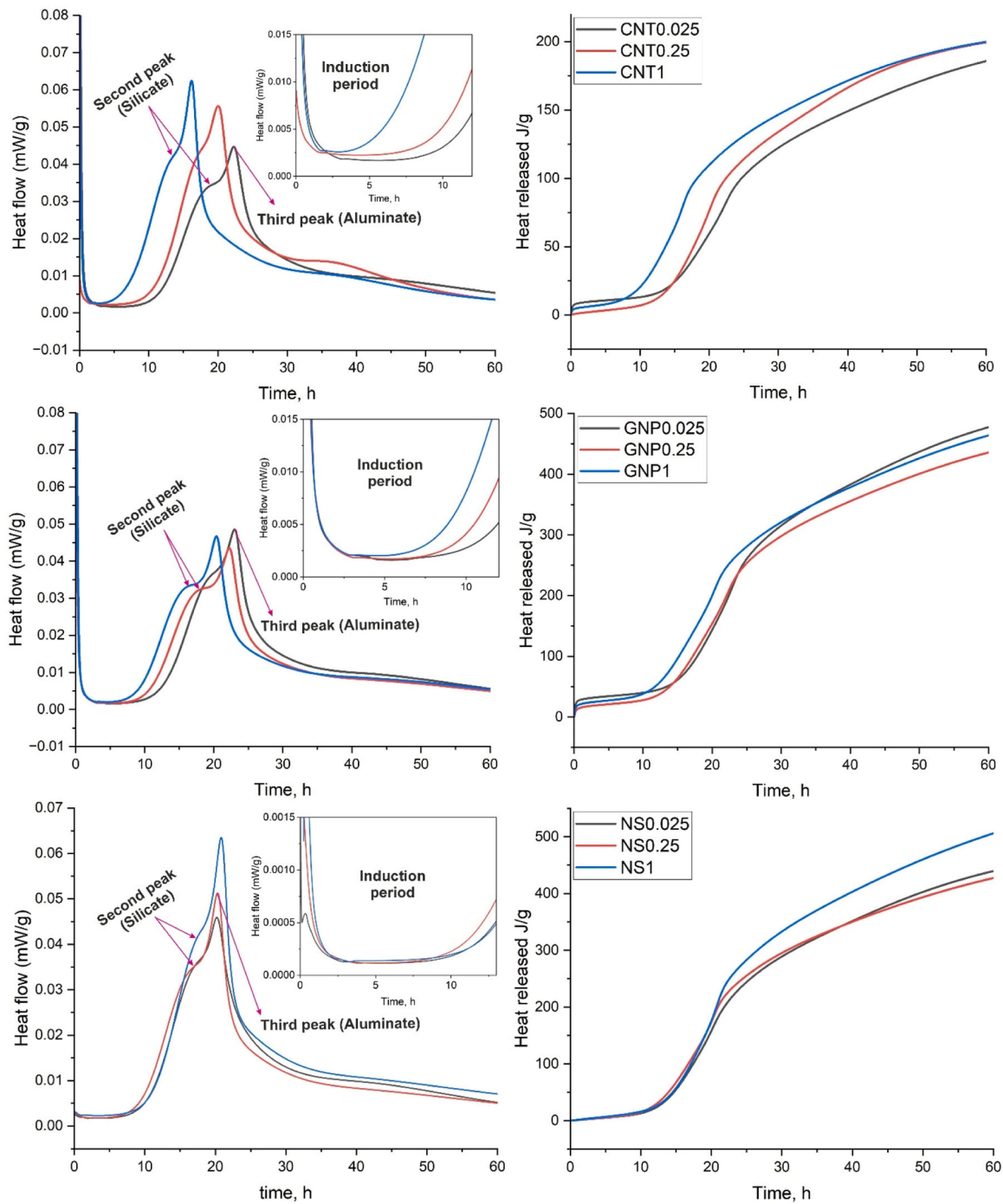


Fig. 4. Heat evolution and cumulative heat release of the incinerated ash mortar with different nano materials.

studied the effect of different lengths of CNTs on the reinforcement efficiency of cementitious material and reported higher reinforcing effect longer CNTs over the shorter ones. In this study probably the higher length of CNT lead to higher reinforcing efficiency and crack filling ability [38]. In addition, TEM analysis also evidences agglomeration of GNP over CNT (refer the specific figure that shows agglomeration). SEM images also evidence the agglomeration of GNP and CNT (refer the specific figure that shows agglomeration), probably due to this phenomenon after certain doses, a sudden drop in strength was observed. The efficiency in strength enhancement for this nano-reinforced mortar was observed to be less than the cement pastes or conventional cement mortars. In a previous study [17], approximately a 42 % improvement in compressive strength was observed in lightweight concrete with 0.6 %

nanotubes, while in this study, only a 29 % enhancement in strength was measured with 0.25 % CNT. This can be explained as, in this study, bottom ash was used which contains 0.026 % metallic aluminium which reacts in alkalis and produces hydrogen resulting in the formation of pores and expansion of cementitious materials. Probably due to this reason, the higher strength was observed for CNT over GNP, attributed to the higher length of CNT.

3.3. Microstructure

SEM images of mortar samples without nanomaterials and with different nanomaterials are shown in the Fig. 6 and Fig. 7. The SEM images presented in Fig. 7 clearly show the presence of pores in the

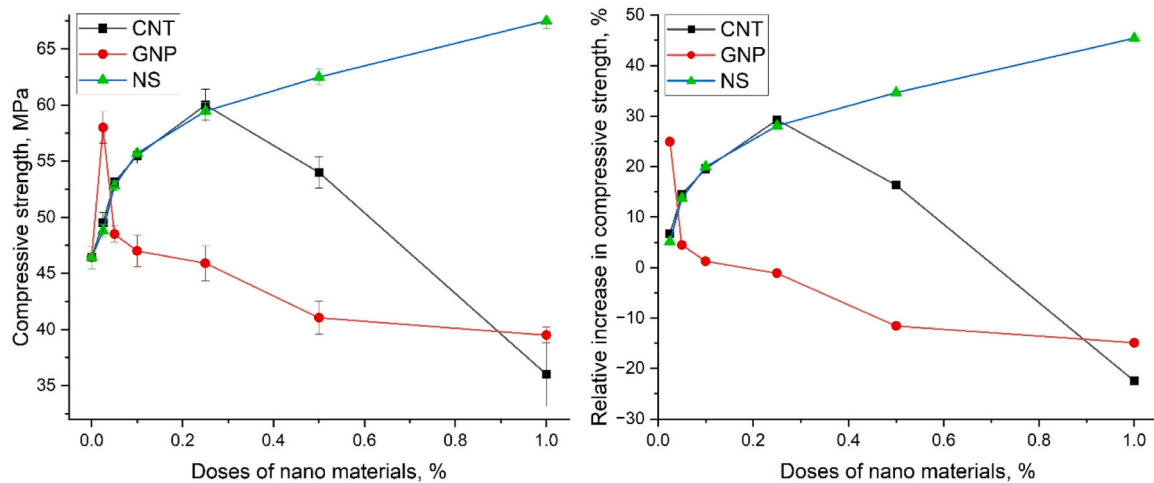


Fig. 5. compressive strength of mortar with different nano materials.

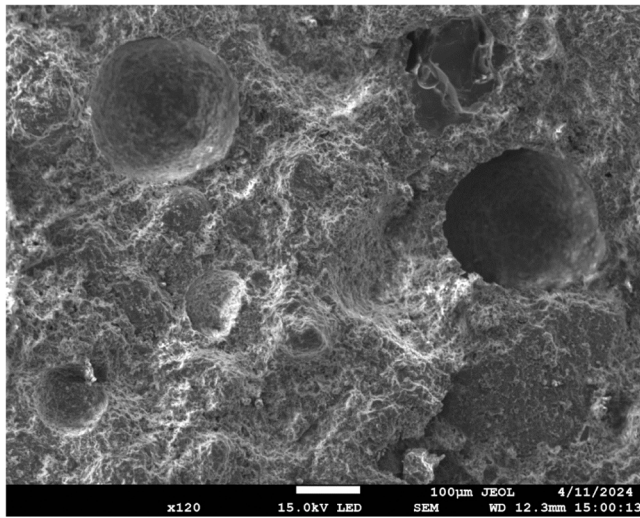


Fig. 6. microscope image of mortar without nano materials.

mortar, which might be attributed to the presence of metallic aluminium in the incinerated ash, leading to the generation of hydrogen gas. SEM images of nano-modified mortar demonstrate a densified microstructure and pore-filling effect. Comparing the microscope images of the incinerated ash mortar, we can clearly observe the variation in appearance at high magnification. At higher doses of CNTs, a network-like distribution was observed within the mortar structure (Fig. 7a). The presence of a high volume of GNP and NS was observed at high magnification, distributing within the mortar matrix. At higher doses of nanomaterials, agglomeration was observed GNP and CNT modified mortars, suggesting that the dispersion method used in this study was not efficient. The dispersion of nanomaterials plays a crucial role in the strength development of cementitious materials. Collins et al. [39] reported that CNTs could be effectively dispersed by ultrasonication energy with polycarboxylate superplasticizer. SEM images of the study also show the hydration products (CSH) surrounding the nanoparticles. The nucleation effect of nanomaterials has been reported by many researchers [17,29,36,38], and it was also observed in this study. Probably due to the pore-filling and nucleation effect of nanoparticles, a higher compressive strength of mortar was observed in this study.

The XRD patterns of nano-modified mortars presented in Fig. 8 show that all of the mortar samples with different nano materials had similar peaks; no additional peaks were observed. Only the intensity of the

mortar samples varied depending on the type of nano materials and their doses. This phenomenon suggests the growth of nearly identical hydration products, with the crystalline of these products influenced by both the type and dosage of nano materials used. Wang and Pang [40], and Adhikary et al. [17,28], also reported similar observations. The thermogravimetric analysis presented in this study was well aligned with this statement. The prominent peaks around 32–34° and 51–52° correspond to the clinker phases Alite (C3S) and Belite (C2S), respectively. The distinct peaks at around 18° and 34°, attributed to Portlandite ($\text{Ca}(\text{OH})_2$), exhibit higher intensities in samples like GNP 0.25, NS 0.25, and NS 0.1, consistent with their enhanced hydration and higher Portlandite content. The peaks around 9° and 16°, characteristic of Ettringite ($\text{Ca}_6(\text{Al}(\text{OH})_6)_2(\text{SO}_4)_3(\text{H}_2\text{O})_{26}$), vary in intensity across samples, reflecting the differences in Ettringite percentages. The peaks associated with Calcite (CaCO_3) and Dolomite ($\text{CaMg}(\text{CO}_3)_2$), around 29° and 39°, show lower intensities in nano-material samples, aligning with their lower carbonate percentages due to potential consumption during hydration reactions. Furthermore, the peaks around 20–30° and 50–60°, attributed to phases like Quartz (SiO_2), Albite ($\text{NaAlSi}_3\text{O}_8$), and Microcline (KAlSi_3O_8), which are present as supplementary cementitious materials (SCMs) or fillers, exhibit varying intensities consistent with the differences in their percentages.

3.4. Thermogravimetric analysis

Thermogravimetric analysis of mortars with nanomaterials, as presented in Fig. 9, and Fig. 10 indicates that all of the samples exhibit almost similar patterns of mass loss. Only the rate of mass loss varies depending on the type of nanomaterials and their concentration. The mass loss in the temperature range of (100–160)°C typically corresponds to the hydration of various cement phases, such as ettringite and calcium silicate hydrate. Further, the weight loss corresponding to the temperature range of (400–520)°C is mainly due to the decomposition of portlandite. Study results show that the incorporation of nanomaterials into the incinerated ash mortar system increased the (ettringite) Aft and C-S-H peaks, as well as the portlandite (CH) peak. The mass loss near 100–160°C for the control sample was measured at about 4.2 %, while with 0.25 % CNT, the mass loss increased to about 5.8 %, and to about 6.1 % for NS and GNP. These results indicate the acceleration of early-age hydration and the generation of a higher content of hydration products. The addition of all the nanomaterials significantly increased the mass loss of portlandite compared to the control sample, suggesting the formation of a higher amount of portlandite in the nano-modified incinerated ash mortar system. The higher amount of portlandite might be attributed to the positive promoting

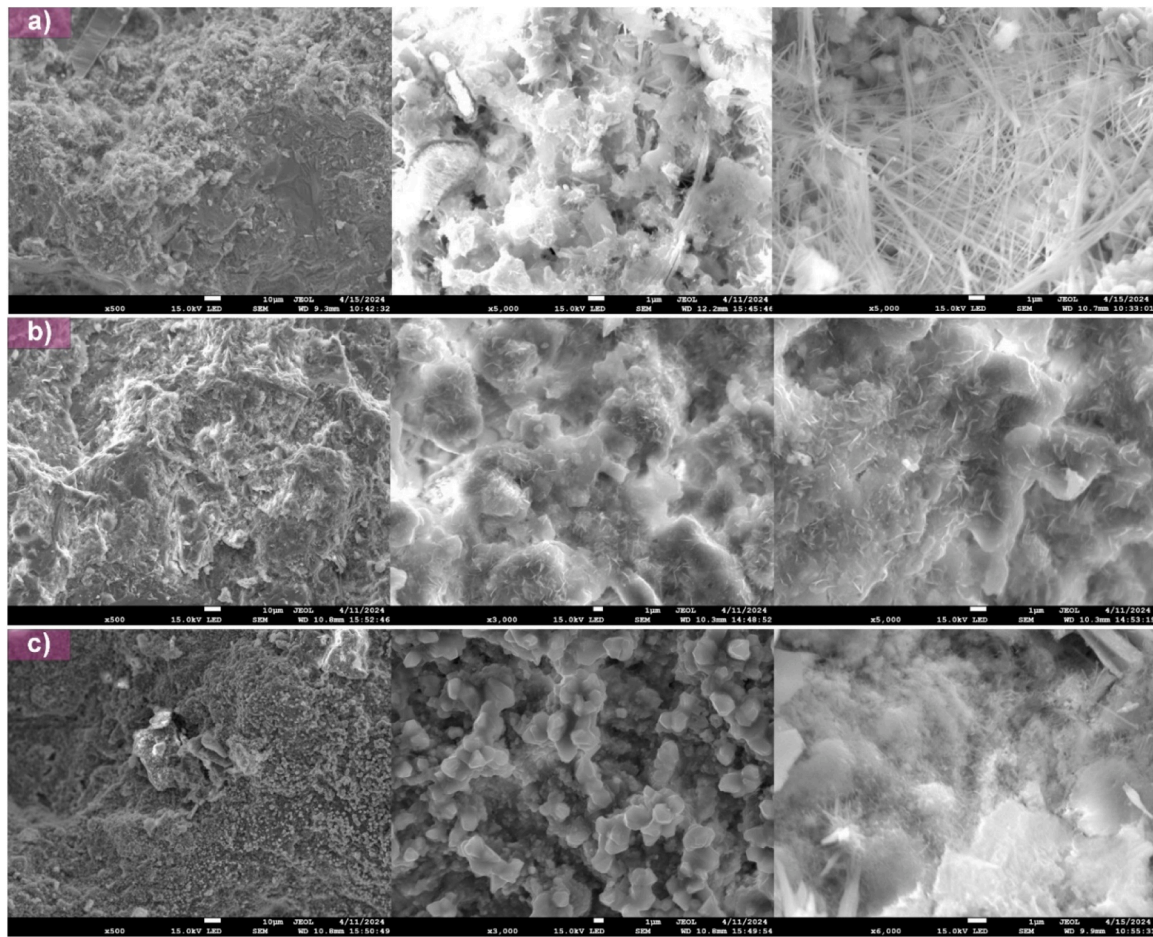


Fig. 7. microscope image of mortar with a) carbon nanotubes; b) graphene nano platelets; and c) nano silica Mineralogical characteristics.

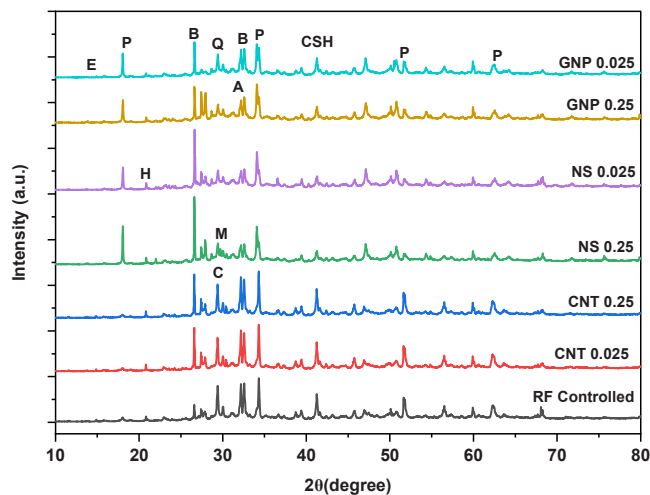


Fig. 8. XRD patterns of mortar with different nano materials. Notes: E- Ettringite, P- Portlandite, H- Anhydrite, B- Belite, A- Alite, Q- Quartz, CSH, C- Calcite, M- Microcline.

effects of nanomaterials. Similar increases in Aft, C-S-H, and CH peaks with nanomaterials were observed by Luo et al. [41]. Comparing the portlandite formation, NS and GNP exhibit a higher amount of portlandite compared to CNT. Despite the acceleration of hydration, the incorporation of CNT in the mortar consumed more portlandite. The weight loss in the temperature range of 740–800°C shows the weight

loss due to the presence of carbonates. The samples with nano materials in this range exhibited lower weight loss compared to the controlled samples, suggesting that carbonates were consumed in the formation of portlandite. Thermogravimetric analysis clearly indicates that mortar samples with nanomaterials hydrated more than the samples without nanomaterials. Thermogravimetric analysis also revealed a higher rate of formation of carbonate phases over conventional mortar. This can be explained by the fact that the samples were kept at room temperature for 14 days before analyzing the thermogravimetric analysis. Bottom ash contains metallic aluminum, which reacts with the alkalis of cement, resulting in the formation of hydrogen gas and an increase in pores of mortar, this could lead to the carbonation of the samples. Interestingly, all the samples with nanomaterials show significantly lower carbonate phases in the mortar compared to the control sample. This might be attributed to the densified structure of the nano-reinforced mortar caused by the pore-filling effect of nanomaterials and the higher growth of hydration products within the mortar [5,29,36,38].

3.5. FTIR analysis

FTIR spectra is conducted on the mortar of PC and PC mixed with nanomaterials at 28 days to provide further characterization, which are shown in Fig. 11. The FTIR results highlight typical groups that can be found in hydrated mortar matrix. For instance, the peak observed at 1113 cm^{-1} are assigned to bending-in-plane vibrations of the Si-O bonds in C₃S [42], two bands at 1430 and 1790 cm^{-1} are attributed to ν_3 and ν_2 C-O stretching of carbonate, respectively [43], and sharp absorption band at 3635 cm^{-1} is the fingerprint of ν_{as} O-H vibration in Portlandite [43]. In addition, the broad bands at 1640 and

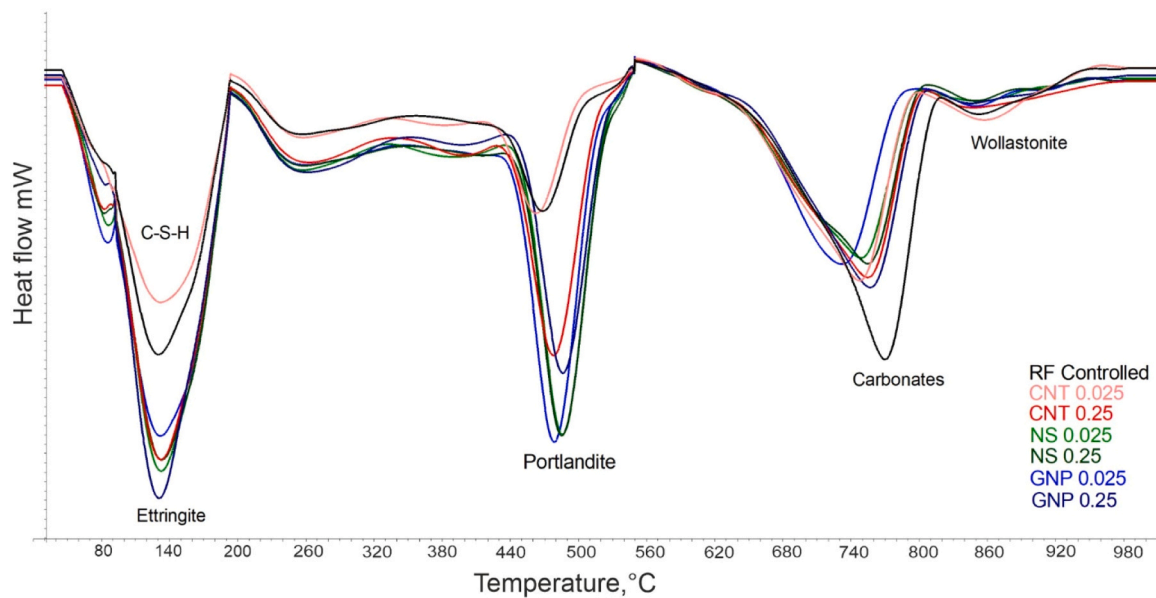


Fig. 9. DTG curves of mortar with different nano materials.

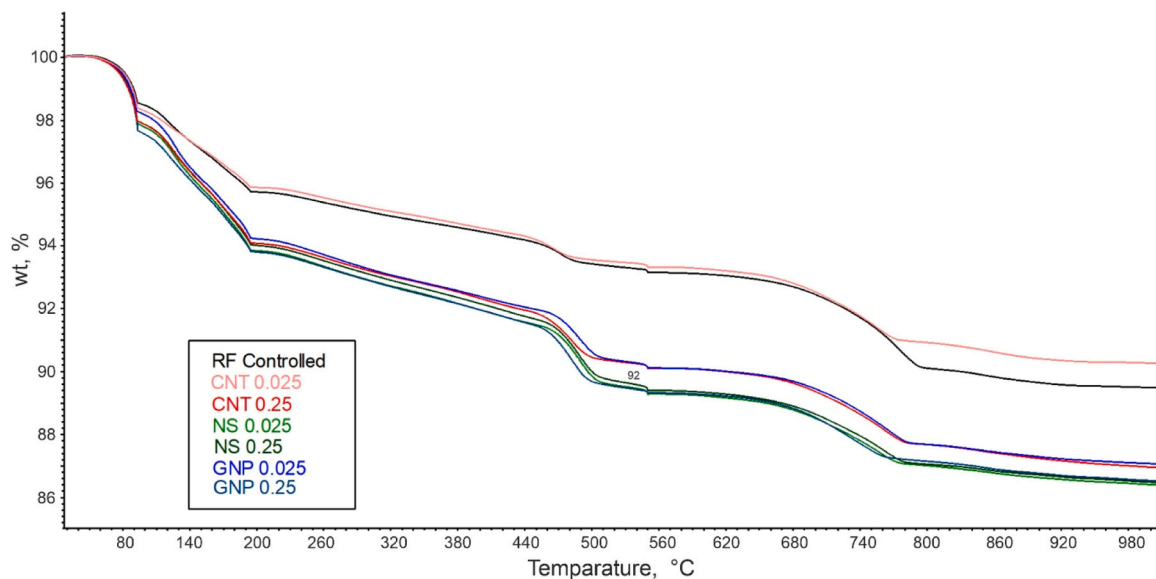


Fig. 10. TGA curves of mortar with different nano materials.

$3400\text{--}3410\text{ cm}^{-1}$ are associated with ν_2 and ν_3 -OH in water and capillary water, respectively [44]. The detected peaks at 1030 cm^{-1} in PC and 0.025 mixed samples are ascribed to Si-O stretch [45]. In the cases of CNT 0.25, GNP 0.25 and NS 0.25, these peaks shifted to $1050\text{--}1060\text{ cm}^{-1}$ that could linked to the appearance of Si-O-Si or Si-O-C network [46]. The observation of band at 954 cm^{-1} in CNT 0.25 and NS 0.025 could indicate Si-O asymmetric stretching (ν_3) of Q^1 and Q^2 species of calcium silicate hydrate (C-S-H) [47]. The peaks ranged $740\text{--}760\text{ cm}^{-1}$ in PC and 0.025 mixed samples could be related to Si-O-Al. Bending vibration of Si-O (SiO_4) and other Si-O groups are found at the band of $450, 568, 581$ and 630 cm^{-1} [48]. Regarding CNT 0.25 and GNP 0.25, the peaks located $676\text{--}688\text{ cm}^{-1}$ could belong to C=O stretching in CNTs and graphene [49].

3.6. Heavy metal leaching

Incinerated ash contains heavy metals that can leach and pollute the

environment. Therefore, a leaching test was conducted to assess the environmental leaching risk. Table 3 indicates that Sb was the only heavy metal in the bottom ash exceeding the threshold limit recommended by Finnish legislation. The study results also suggested that Sb was stabilized in the cement binding system, and the leaching of all the heavy metals listed in Table 3 was encapsulated and leached far below that of the raw bottom ash. It was observed that incorporating nano-materials did not significantly impact the leaching of heavy metals. Incorporating GNP and NS showed a negligible increase in Cr leaching, while a negligible decrease in Mo was measured. The leaching of Sb, V, and As remained unchanged with different nano materials. Due to the very low heavy metal content in the bottom ash and the cementitious system being a well-known binder for effectively stabilizing heavy metals, the effect of different nanomaterials on heavy metal leaching cannot be conclusively concluded.

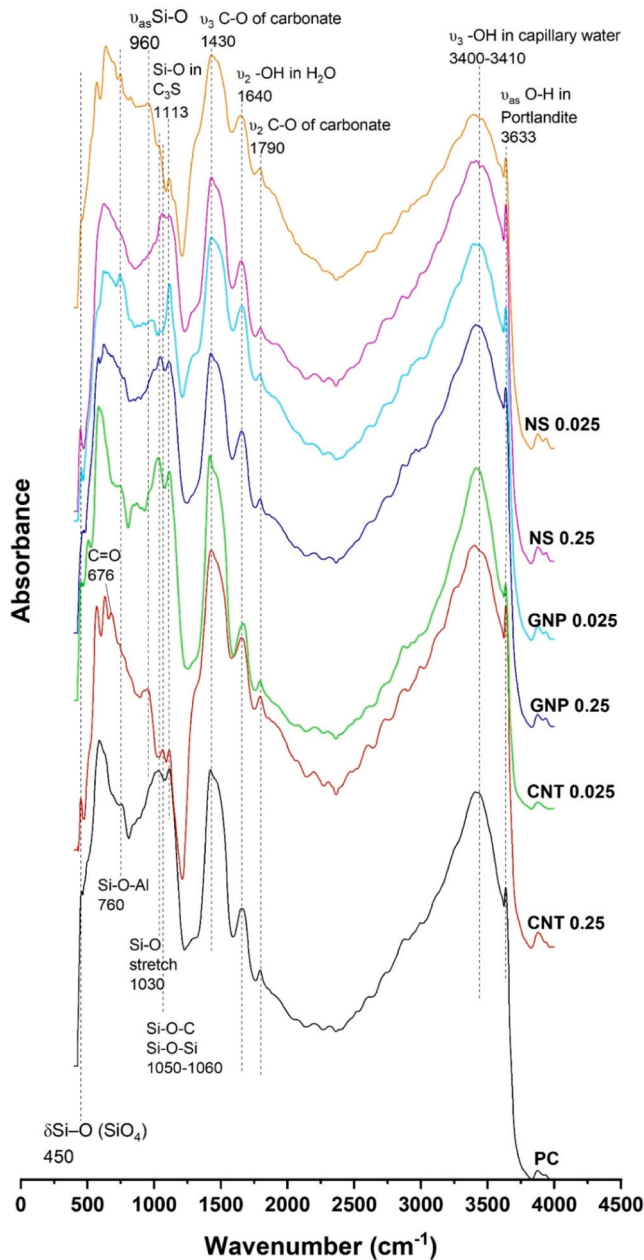


Fig. 11. FTIR spectra of mortar with different nano materials.

3.7. Environmental sustainability assessment

Fig. 12. illustrates the comparative global warming potential in terms of kg CO₂ eq. compared to compressive strength (MPa) of mortar for different doses of nanomaterials (%). A similar inference is observed for other impact categories as well, however, to avoid repeatability only global warming potential is discussed. It is observed that with the

increase in doses of nanomaterials the emissions increase, which is obvious considering the impact of nanomaterials keeping the other processes constant. Based on the emissions aspects GNP is the most sustainable, followed by NS, and comparatively, CNT is the least sustainable. The sustainable zones for GNP and NS are marked in Fig. 12 indicating that up to 0.9 % dose of GNP the mortar can be considered sustainable, however, further increase in the dose is not sustainable considering the achieved compressive strength aspect. The compressive strength was achieved at 0.025 % GNP dose is the maximum with the least carbon emission. For NS the trend for compressive strength concerning the associated emissions will continue to be more sustainable as the gap between increases with the dose. In contrast to GNP and NS, CNT emits more, and less compressive strength is achieved. Overall, considering all the mixes it is concluded that for lower doses GNP is the most sustainable and for higher doses NS is the most sustainable. It is important to mention here that these observations are derived considering information provided from the literature stated in the methods section [1–5] and are very limited. Further, studies in this regard are needed for more precise conclusions as with the advancement in technology lower emissions can be achieved for manufacturing nanomaterials.

4. Conclusions

This study provides a comprehensive comparative analysis of the effects of different types of nanomaterials on the technical properties of incinerated ash mortar. Based on the obtained results, the following conclusions can be drawn.

- The optimal strength gain of incinerated ash mortar incorporating nanomaterials varied depending on the type of nanomaterial. NS-incorporated ash mortar showed stronger compressive strength than those with CNTs and GNPs. Nano silica-incorporated incinerated ash mortar achieved optimal strength at a 1 % dose, resulting in

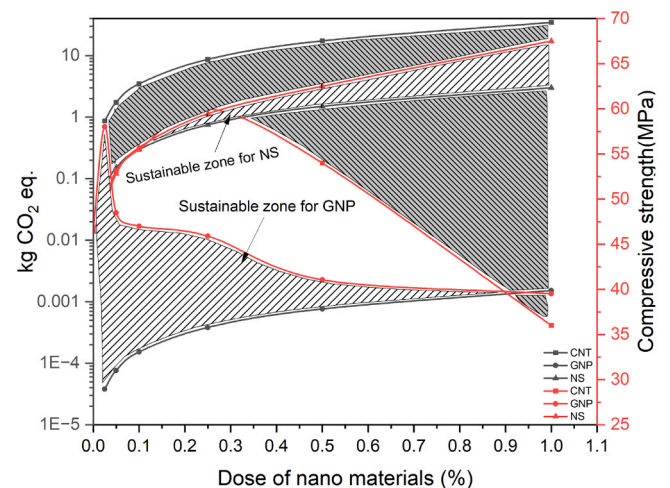


Fig. 12. Comparative global warming potential (kg CO₂ eq.) with respect to compressive strength at different doses of nano materials.

Table 3

Heavy metal leaching from nano modified incinerated ash mortar.

Elements, mg/kg	Raw bottom ash	Control sample	GNP 0.025	CNT 0.25	NS-1	Finnish Legislation limit
As	0.14	<0.01	<0.01	<0.01	<0.01	2
Cr	0.45	0.083	0.091	0.069	0.096	10
Mo	0.3	0.069	0.065	0.079	0.06	6
Pb	0.015	0.02	0.031	0.014	0.011	2
Sb	0.76	<0.01	<0.01	<0.01	<0.01	0.7
V	0.26	<0.01	<0.01	<0.01	<0.01	3

a 45 % improvement in compressive strength. In contrast, the optimal strength gain for incinerated ash mortar with CNTs and GNPs was observed at doses of 0.25 % and 0.025 %, with enhancements of 29 % and 24 % in compressive strength, respectively.

- The induction period, C3S, and aluminate hydration of the incinerated ash mortar were accelerated with the increase in nanomaterial doses. However, a 1 % CNT addition showed the most accelerated induction period among GNPs and NS. The effectiveness of silicate and aluminate hydration was more significant with NS and CNTs compared to graphene nanoplatelets.
- SEM images show the reinforcing effect of nanomaterials, and hydration products surrounding the nanomaterials demonstrate the nucleation effect of nanomaterials. Thermogravimetric analysis and calorimetry results also supports the nucleation effect of nanomaterials.
- GNP at 0.025 % dose is found to be the most sustainable in the present study. However, with an increased dose the NS is more environmentally friendly over GNP and CNT considering the compressive strength aspect. Due to the lack of robust inventory future studies in this regard are necessary for reliable conclusions.

CRedit authorship contribution statement

Michael Antwi: Writing – original draft, Methodology, Formal analysis, Data curation. **Jitendra Patel:** Writing – original draft, Methodology, Formal analysis, Data curation. **Suman Adhikary:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Anshumali Mishra:** Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Priyadharshini Perumal:** Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Funding acquisition, Formal analysis, Data curation. **Yangmei Yu:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tuan Truong:** Writing – original draft, Validation, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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